

UNIVERSITY OF BATH • SEPTEMBER 2026

Formalization and physics: a new frontier for AI



Joseph Tooby-Smith

Mathematical Foundations group, Department of Computer Science



[leanprover-community/physlib](https://leanprover-community.github.io/physlib)



josephtoobysmith.com

A new era of proving

MAY 2026

The unit-distance problem

Erdős, 1946. Disproved.

OpenAI, internal model

JUL 2026

The Jacobian conjecture

Keller, 1939. Counterexample in $\mathbb{C}^3$.

Alpoge, with Claude Fable 5

AUG 2026

Ten open problems

From maths and CS

OpenAI “Astra”

SEP 2026

Navier–Stokes existence

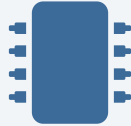
Disproved.

OpenAI, internal model

Two ways of verifying



Expert review



Computer

The Navier–Stokes announcement

The screenshot shows the OpenAI website with a navigation bar at the top containing links for Research, Products, Business, Developers, Company, and Foundation, along with a search icon. On the right side of the navigation bar are buttons for 'Log in' and 'Try ChatGPT'. The main content area features a date 'September 8, 2026' and the categories 'Research' and 'Publication'. The title of the announcement is 'On the Navier–Stokes Millennium Prize Problem'. Below the title are two buttons: 'Read the paper' and 'Link to Lean formalized proof >'. Further down, there is a section for audio content with a play button icon, the text 'Listen to article', a duration of '8:43', and a 'Share' button with a share icon. At the bottom left, there are two sub-sections: 'The problem' and 'The result'. The main text of the announcement states: 'We're sharing a solution to the Navier–Stokes existence and smoothness problem, one of the Millennium Prize Problems. This proof, produced by an internal OpenAI'.

OpenAI Research Products Business Developers Company Foundation 🔍

Log in Try ChatGPT ↗

September 8, 2026 Research Publication

On the Navier–Stokes Millennium Prize Problem

Read the paper Link to Lean formalized proof >

▶ Listen to article 8:43 Share

The problem The result

We're sharing a solution to the Navier–Stokes existence and smoothness problem, one of the Millennium Prize Problems. This proof, produced by an internal OpenAI

A rapid overview of Lean

One way to check by computer

Definitions, theorems, proofs



consistency & correctness, guaranteed

Built on type theory

Lean knows the **type** of everything, and checks that things have the type expected.

$\underbrace{p}_{\text{proof}} : \underbrace{P}_{\text{theorem}} \quad \underbrace{7}_{\text{term}} : \underbrace{\mathbb{N}}_{\text{type}}$

A definition, a theorem, a proof

```
def energy (m v : Real) : Real :=  
  m * v ^ 2 / 2
```

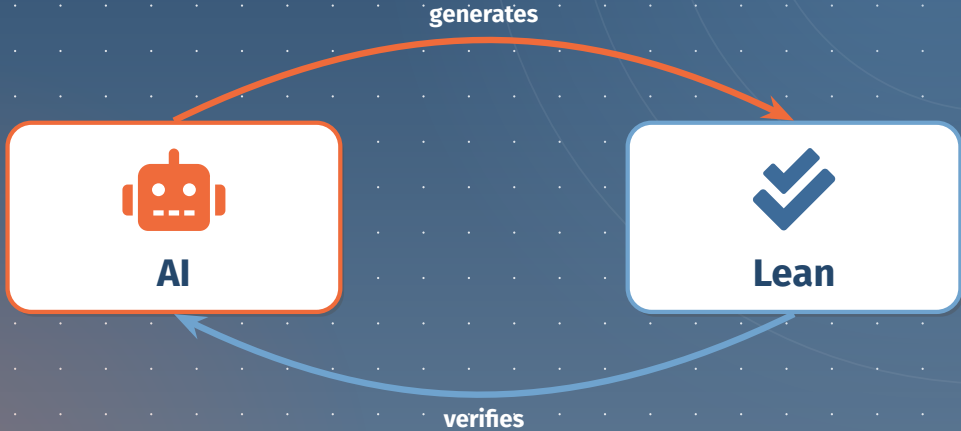
```
theorem energy_nonneg  
  (m v : Real) (hm : 0 <= m) :  
  0 <= energy m v := by  
  unfold energy  
  positivity
```



 Open the live editor

scan to open on your phone

A feedback loop



What the loop is built on



leanprover-community / mathlib4

A *monolithic* library of mathematics formalized in Lean — large enough to work at a high level.

public

Lean 4

● Lean 📄 100,000 definitions ✓ 200,000 theorems 👤 500+ contributors ⌚ started 2017

Formalizing Fermat's Last Theorem

Formalizing Fermat's Last Theorem in Lean

A timeline and selected excerpts from Claude's reasoning

Anthropic

Abstract

We provide some background on Claude's formalization of Fermat's Last Theorem in Lean, including a timeline of the proof and verbatim excerpts from its reasoning. The proof took about eleven days and proved Fermat's Last Theorem relying only on Lean's three standard axioms, with no unproved placeholders (sorry). The full proof can be found at github.com/anthropics/fermats-last-theorem.

My broad interest



leanprover-community / physlib

A community-driven project to digitalise results from physics into Lean.

public

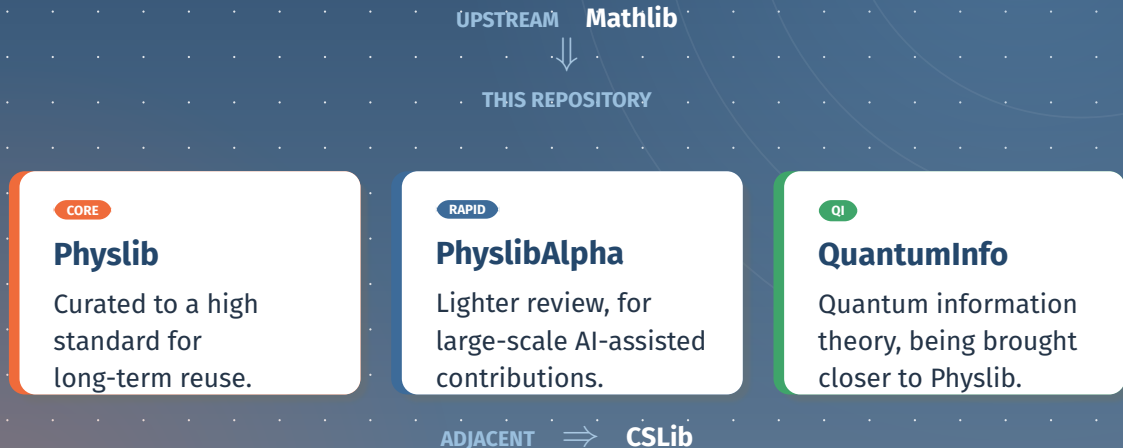
Lean 4

● Lean ★ 676 stars 🔗 153 forks 👤 85 contributors ⌚ started 2024

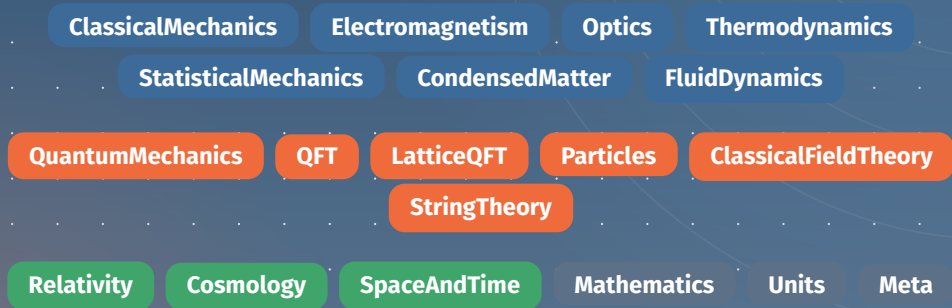
What Physlib is

- A **monolithic library** for physics results from standard physics (not just axiomatisations).
- Research-level results, with carefully standardized definitions and theorems — a foundation for future formalization.
- Community run (no bias to any academic or company), following open-source software workflows.

Structure of Physlib



What Physlib contains



Interesting results in Physlib

- ✓ Classical harmonic oscillator
- ✓ Quantum harmonic oscillator
- ✓ Maxwell's equation
- ✓ Lorentz group
- ✓ Twin paradox
- ✓ Tight binding chain
- ✓ Anomaly cancellation
- ✓ CKM matrix
- ✓ Wick's theorem
- ✓ String theoretic

 Example of Higgs result

Ongoing work

$$\begin{aligned}\mathcal{L}_{\text{SM}} = & -\frac{1}{4} G_{\mu\nu}^A G^{A\mu\nu} - \frac{1}{4} W_{\mu\nu}^I W^{I\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu} + \sum_{\psi} i \bar{\psi} \gamma^{\mu} D_{\mu} \psi \\ & + (D_{\mu} H)^{\dagger} (D^{\mu} H) + \mu^2 H^{\dagger} H - \lambda (H^{\dagger} H)^2 \\ & - \left(\bar{Q}_L Y_u \tilde{H} u_R + \bar{Q}_L Y_d H d_R + \bar{L}_L Y_e H e_R + \text{h.c.} \right)\end{aligned}$$

$$\begin{aligned}D_{\mu} &= \partial_{\mu} - ig_3 G_{\mu}^A T^A - ig_2 W_{\mu}^I t^I - ig_1 Y B_{\mu}, & B_{\mu\nu} &= \partial_{\mu} B_{\nu} - \partial_{\nu} B_{\mu}, \\ G_{\mu\nu}^A &= \partial_{\mu} G_{\nu}^A - \partial_{\nu} G_{\mu}^A + g_3 f^{ABC} G_{\mu}^B G_{\nu}^C, & W_{\mu\nu}^I &= \partial_{\mu} W_{\nu}^I - \partial_{\nu} W_{\mu}^I + g_2 \epsilon^{IJK} W_{\mu}^J W_{\nu}^K.\end{aligned}$$

$G = SU(3) \times SU(2) \times U(1)$, acting on three generations of $Q_L(\mathbf{3}, \mathbf{2})_{1/6}$, $u_R(\mathbf{3}, \mathbf{1})_{2/3}$, $d_R(\mathbf{3}, \mathbf{1})_{-1/3}$, $L_L(\mathbf{1}, \mathbf{2})_{-1/2}$, $e_R(\mathbf{1}, \mathbf{1})_{-1}$, and $H(\mathbf{1}, \mathbf{2})_{1/2}$.

Other work on Lean at Bath

Proof mining in Lean

A library for partially automating the extraction of computational information from proofs, with a focus on convergence rates for optimization algorithms.

Thomas Powell • Ben Langton

Thank you for watching



Joseph Tooby-Smith • University of Bath



`leanprover-community/physlib`



`josephtoobysmith.com`